Giung Lee

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Research Interests

Robot Safety, Safe Learning-Based Robotics, Risk-Aware Planning, Visual Navigation, Robot Manipulation, and Multi-Robot Systems

Education

Rice University, George R. Brown School of Engineering and Computing

- *Bachelor of Science in Computer Science, Minor in Statistics*
- Anticipated Graduation Date: May 2027
- Cumulative GPA: 3.77/4.0

Publications & Manuscripts

- **Giung Lee**, Weihang Guo, and Lydia E. Kavraki. *Foundation-Model-Guided Topology-Aware Semantic Risk Fields for Manipulation*. Under review at ICRA 2027. [arXiv]
- Geonyeong Ko[†], **Giung Lee**[†], and Changjoo Nam. *SAFER-Nav: Enhancing Safety for Visual Robot Navigation via Segmentation-Aware Fine-Tuning*. Under review at ICRA 2027. [arXiv] [Project Page]

[†] Equal contribution.

Research Experience

Kavraki Lab, Rice University – Houston, TX Jan 2026 – Present

Undergraduate Researcher | Advisor: Prof. Lydia E. Kavraki

- Developed a **3D semantic risk field for robot manipulation** to represent physically feasible motions that are **collision-free but unsafe in context**.
- Designed an **object-conditioned, topology-aware 3D voxel field** whose risk distribution changes with the manipulated object. It combines foundation-model-derived directional risk priors with 3D scene geometry, geodesic spatial decay, and **topology-aware shielding based on the Euclidean-to-geodesic distance ratio**.
- **Led framework development and evaluation**, including GPU-based field construction, risk-aware manipulation planning, systematic counterfactual and component ablations, quantitative analysis, figure design, and primary manuscript drafting.
- Evaluated risk-aware trajectories against independent human-authored reference risk fields, reducing **mean path risk by 74.1% on average** across three household manipulation scenarios relative to collision-only planning under identical geometric constraints; further validated topology-aware shielding across vertical, lateral, and multi-surface barriers.

AI Robotics Lab, Sogang University – Seoul, South Korea Jun 2025 – May 2026

Undergraduate Researcher | Advisor: Prof. Changjoo Nam

- Developed a **segmentation-aware approach to visual robot navigation** that trains the policy to **represent obstacles and traversable free space**, rather than relying solely on post-hoc trajectory correction or geometric safety priors.
- Designed a **segmentation-aware fine-tuning architecture** that incorporates traversability features into a frozen pretrained RGB navigation backbone through a **residual attention bridge and an auxiliary segmentation-only action branch**, allowing the policy to predict collision-aware waypoints while preserving goal-conditioned navigation.
- **Initiated the project by defining the research problem** and helping establish segmentation as an internal safety representation; co-led real-world experiments and contributed extensively to manuscript drafting and figure design.
- Evaluated SAFER-Nav across **three robot platforms and two unseen indoor environments**, including static- and dynamic-obstacle tests, against ViNT, NoMaD, and their CARE-augmented variants.
- Across static-obstacle tests, achieved **90%–100% goal arrival** with **0–0.4 collisions per run**, outperforming ViNT, NoMaD, and their CARE-augmented variants in goal arrival and collision frequency; across 20 dynamic-obstacle trials, SAFER-Nav remained **collision-free in all 20 trials**, compared with collisions in 14 trials for each of ViNT and NoMaD and 4–5 trials for the CARE-augmented baselines.

Robotics Lab, Texas A&M University – Remote Jun 2026 – Present
Undergraduate Researcher | Advisor: Prof. Yiwei Lyu

- Developed a **safety intervention framework for frozen Vision-Language-Action (VLA) policies** that introduces geometry-derived physical safety references into the policy's generative process without fine-tuning the pretrained model.
- Designed a **reverse-flow latent steering method** that maps physical safety references into the VLA's latent flow source space and regenerates policy-native actions through the original frozen model.
- Developing an **RRT-based geometric recovery planner** for VLA safety dead-ends, generating collision-free recovery motions toward higher-clearance states to support safe handoff back to the frozen policy.

Selected Experience

Teaching Assistant | Rice University

- **COMP 450: Algorithmic and AI-Enabled Robotics** Aug 2026 – Present
- **STAT 310: Probability and Statistics** Aug 2026 – Present
- **COMP 215: Introduction to Program Design** Aug 2025 – May 2026

UAS Mission Commander (Sergeant) | U.S. Army (KATUSA) – South Korea Nov 2022 – May 2024

- Supervised 300+ UAV flight operations as mission commander and squad leader, coordinating safety checks, mission execution, and U.S.–ROK training requirements.

SPACE-K | UAV Tracking and Communication System Mar 2023 – Apr 2024

- Led a student team as a group leader in a UAV business competition project replacing LTE-based communication with SATCOM; earned 1st Place at the 2024 KT&G Alliance Students' Venture Forum and a Creativity Award at the Army Start-up Business Competition.

Honors & Awards

- **Korea Presidential Science Scholarship** | 1 of 20 students selected nationwide
- **Rice University President's Honor Roll** | Spring 2026, Fall 2025
- **1st Place** | 2024 KT&G Alliance Students' Venture Forum
- **Creativity Award** | 2024 Army Start-up Business Competition

Technical Skills

- **Programming:** Python, C/C++, Java, R
- **Machine Learning & Generative Modeling:** PyTorch, JAX, multimodal transformers, flow matching, vision-language-action (VLA) inference and latent steering, semantic segmentation
- **Robotics & Planning:** ROS 2, MuJoCo, PyRoki, OMPL, motion planning, robot learning, real-world robot deployment
- **Languages:** English, Korean